



Allergic Reaction / Anaphylaxis

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA Tape.
- Fluids and medications titrated to maintain SBP > 70 + (age x 2) mmHg

Signs & Symptoms:

- Edema / Voice Changes
- Itching or Hives
- Coughing / Wheezing or Respiratory Distress
- Chest or Throat Constriction / Tightness
- Difficulty Swallowing
- Hypotension or Shock
- Vomiting / Diarrhea

Differential:

- Urticaria (rash only)
- Anaphylaxis (systemic effect)
- Shock (vascular effect)
- Angioedema (drug induced)
- Aspiration / Airway Obstruction
- Vasovagal event
- CHF
- Asthma or COPD
- Anxiety

Clinical Management Options

P	P	P	P	P	P	• Assist with Patient's Epinephrine delivery device for severe respiratory distress and/or anaphylaxis. • If the patient does not have Epinephrine, then Epinephrine • Oxygen, target SpO₂ 92 – 96% • Cold pack to insect bite or sting site and remove bee stinger if present • Basic airway management as needed
L	L	L	L	L	L	• Albuterol • Diphenhydramine – Adult PO only • CPAP, if refractory to albuterol • Monitor ETCO₂
1	2	3	4	5	6	• IV / IO access as appropriate for patient condition • IV fluid therapy with isotonic crystalloid, titrated to Adult SBP ≥ 90 mmHg or MAP >65 mmHg • Diphenhydramine • Dystonic Reaction: Diphenhydramine
						• Epinephrine, up to 3 additional doses • Dexamethasone • Assess ECG • Push dose Epinephrine for hypotension • Epinephrine infusion until the patient stabilizes • Advanced Airway Management as needed
						• Rocuronium for Rapid Sequence Induction • Use Airway Management Checklist for all Rapid Sequence Inductions

Consult Online Medical Control As Needed

Pearls:

- **Anaphylaxis is defined as acute multiorgan (≥2) system reaction. Common systems involved include: respiratory, cardiovascular, cutaneous, and gastrointestinal (GI).**
- Anaphylaxis is extremely undertreated in pediatric patients and aggressive anaphylaxis treatment is encouraged.
- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- Continuous reassessment for rebound reaction with need for additional epinephrine.
- Lung sounds should be assessed between each dose of Albuterol prior to additional nebulizers.
- Any patient with respiratory symptoms or extensive reaction should receive IV/IO or IM diphenhydramine.
- The shorter the onset from exposure to symptoms, the worse the reaction.
- Epinephrine is the single most important intervention in this setting and has small risk for high benefit.